

# Ionanofluids based on ionic liquid mixtures, a new approach as an alternative material for solar energy storage

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## ABSTRACT

Hybrid materials based on ionic liquids (ILs) known as Ionanofluids (INFs) have shown exciting improvements in their thermophysical properties compared to their base fluids (BF). Within this group, INFs obtained using mixtures of ionic liquids as BFs have been virtually ignored until now. In this work, INFs composed of two equimolar mixtures of ILs, [Omim][PF<sub>6</sub>] + [Odmim][PF<sub>6</sub>] and [Omim][PF<sub>6</sub>] + [Odmim][BF<sub>4</sub>] as base fluids and Multiwalled Carbon Nanotubes (MWCNTs) with weight percentages of 0.04, 0.06, 0.08, and 0.1 wt% were elaborated. Furthermore, the thermophysical properties of our proposed new materials were evaluated and compared to conventional materials currently used in solar energy storage systems. It was shown that the heat capacity ( $C_p$ ) reaches increases of up to 3.7 and 3.2 times that of nitrated salts and the commercial fluid Therminol VP-1, which translates into a 9.4% and 284% higher thermal storage density ( $E$ ), respectively, and even lower thermal diffusivity ( $\alpha$ ), which supports the preliminary idea of using these new materials for energy storage.

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## 1. Introduction

The progress in renewable energy resources has proven that the most significant challenge is developing alternative storage and energy production approaches. In this respect, Concentrated Solar Power (CSP) is one of the technologies of most significant interest today due to its high capacity to generate electricity from solar energy. CSP plants consist of a group of solar collector mirrors that reflect the sun's rays to a receiving zone where the radiation is concentrated in a Heat Transfer Fluid (HTF) or stored in a Thermal Energy Storage (TES) system [1].

The primary materials used as HTF and TES in CSP plants are nitrate salts (40% KNO<sub>3</sub> + 60% NaNO<sub>3</sub>), eutectic biphenyl, diphenyl oxide (therminol VP-1), and mineral or silicone oil. These materials have limitations in their application. Nitrated salts cause corrosion and have a high melting point, which risks solidification in cold seasons. Therminol VP-1 has a high vapor pressure, which could damage pipes at high temperatures and increase the risk of ignition. Mineral oils and silicones have a limited working range because they decompose at low temperatures [2]. For this reason, research into innovative materials with advanced properties is a constant concern.

Traditionally, ionic liquids (ILs) have been considered as heat transfer and thermal storage fluids due to their unique properties such as chemical and thermal stability, low vapor pressure, non-flammability, and melting points generally < 373.15 K with a wide temperature operating range in the liquid state [3].

The same approach used with nanoparticles to improve the thermophysical properties of fluids generating a nanofluid [4] has been used with ionic fluids, where the addition of thermally conductive nanoparticles originates Ionanofluids (INFs); concept incorporated by Nieto de Castro et al. [5].

In the last decade, a considerable amount of literature has been reported on INFs. Among them, INFs based on alkyl imidazolium and phosphonium as base fluid (BF) in combination with carbon nanotubes (CNTs), [6,7] have shown to impact important thermophysical properties such as thermal conductivity, specific heat, density, and viscosity.

Recently, Das et al. [8] did a comprehensive review about modified ionic liquids and ionanofluids with nanomaterials for solar energy applications. Among their findings, ILs have higher heat capacities, and the addition of nanomaterials increases the storage density of base fluids. [9] Moreover, small additions of nanomaterials such as graphene [10], MXene [11], and MWCNTs [12] improve the absorbance of ILs.

Despite the number of authors who have empirically evaluated, simulated, or analyzed theories on the properties of INFs, research

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continues improving thermal properties. The thermal conductivity and specific heat are influenced by the concentration and type of nanomaterial, nature of the base fluid, and temperature. These variables are debated in each new research.

On the other hand, many researchers [13–18] have studied thermophysical properties of ionic liquid mixtures such as density, viscosity, and heat capacity.

Studies carried out by this group demonstrated before showed those certain binary mixtures of 1-octyl-3-methylimidazolium and 1-octyl-2, 3-dimethyl imidazolium-based ionic liquids (Fig. 1) increased their heat capacity and, therefore, their thermal storage density, enabling them to be primarily compared with materials conventionally used in solar energy storage [19].

The novelty of this study resides in the preparation of INFs using mixtures of ILs as a base fluid. These new materials have not been fully explored, and since three or more components form them, the possibility of enhancing properties between them increases. In this work, INFs were prepared using an equimolar mixture of [Omim][PF<sub>6</sub>] + [Odmim][PF<sub>6</sub>] (ILM1) and [Omim][PF<sub>6</sub>] + [Odmim][BF<sub>4</sub>] (ILM2) as base fluid and MWCNTs. In addition, these hybrid materials were fully characterized as a function of MWCNTs weight fraction (0.04 to 0.1 wt%) and temperature. Properties such as heat capacity, thermal conductivity, density, viscosity, thermal storage density, and thermal diffusivity were compared with conventional materials used in CSP plants.

## 2. Materials and methods

### 2.1. General information

All reagents were used as purchased from commercial sources without further purification. 1-methylimidazole (C<sub>4</sub>H<sub>6</sub>N<sub>2</sub>, Merck Millipore, >99.0%), 1,2-Dimethylimidazole (C<sub>5</sub>H<sub>8</sub>N<sub>2</sub>, Merck Millipore, >98.0%), 1-Bromooctane (CH<sub>3</sub>(CH<sub>2</sub>)<sub>7</sub>Br, Sigma Aldrich, 99.0%), Potassium Hexafluorophosphate (KPF<sub>6</sub>, Sigma Aldrich, ≥99.99%), Sodium tetrafluoroborate (NaBF<sub>4</sub>, Sigma Aldrich, ≥98%), acetone (CH<sub>3</sub>COCH<sub>3</sub>, EMSURE® Merck Millipore, ≥99.8%), acetonitrile (CH<sub>3</sub>CN, Merck Millipore, ≥99.8%), chloroform (CH<sub>3</sub>Cl, EMSURE® Merck Millipore, ≥99.8%), dichloromethane (CH<sub>2</sub>Cl<sub>2</sub>, EMSURE® Merck Millipore, ≥99.8%), ethyl acetate (C<sub>4</sub>H<sub>8</sub>O<sub>2</sub>, EMSURE, Merck Millipore, ≥99.5%), sodium sulfate anhydrous (Na<sub>2</sub>SO<sub>4</sub>, EMSURE® Merck Millipore, ≥99.0%), diatomaceous earth (Kieselguhr, GR for analysis, Carbon nanotube, multi-walled thin and short CAS number: 308068–56-6 < 5% Metal Oxide (Sigma-Aldrich, <5% Metal Oxide), Merck Millipore), Milli-Q distilled water was used in all the syntheses.

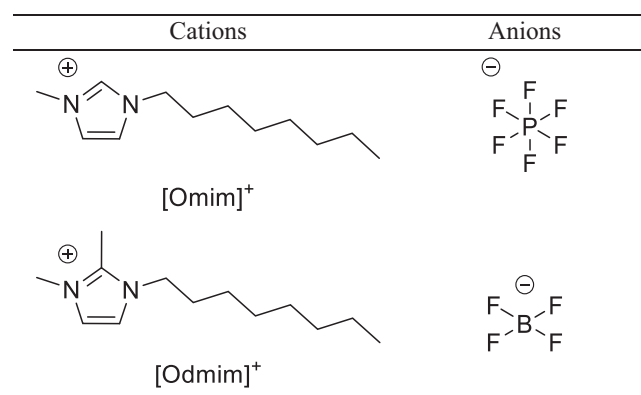


Fig. 1. Structure of ionic liquids used as base fluids.

### 2.2. Experimental section

The ionic liquids used in this study were synthesized as reported in the literature and are described extensively in the [Supplementary material](#). Briefly, [Omim][Br] and [Odmim][Br] were obtained by reaction of 1-methylimidazole (or 1,2-dimethylimidazole) with an excess of 1-bromooctane. Then, ionic liquids were prepared by the metathesis reaction of the corresponding bromide with sodium tetra fluoroborate or potassium hexafluorophosphate.

<sup>1</sup>H and <sup>13</sup>C NMR spectra were recorded on Bruker AMX-400 (11.74 T, 400 MHz to <sup>1</sup>H, and 126 MHz to <sup>13</sup>C) NMR spectrometers using the residual proton or the carbon signal of the deuterated solvent as an internal standard.

The thermal stability of Ionanofluids is given by their base fluids, which TGA evaluated in a previous job. [19] [Omim][PF<sub>6</sub>] + [-Odmim][PF<sub>6</sub>] and [Omim][PF<sub>6</sub>] + [Odmim][BF<sub>4</sub>], registered a decomposition temperature (Tonset) of 677.15 K and 675.15 K respectively.

Differential Scanning Calorimetry (DSC) was carried out using a Perkin Elmer DSC 6000; employing nitrogen as an inert gas at 20 ml/min., Samples were hermetically sealed in standard 40 μL aluminum capsules using an empty one as reference. The analysis was performed by dynamic flow heating from 298.15 K to 473.15 K at a heating rate of 10 K/min with 2 min of isotherm at the beginning and 1 min at the end of each analysis. Calibration for temperature and enthalpy measurements was performed using certified indium and zinc standards as a reference, with an uncertainty of ± 0.2 K for temperature measurement and 2% for enthalpy change. The heat capacity was obtained using Pyris manager software and determined a sample of sapphire used as a standard. As a comparison test, the procedure was applied to the base fluids, and the results were consistent with Cp values previously reported in the literature.

Thermal conductivity was obtained using a Decagon Device thermal properties analyzer, model KD2 Pro, with the KS-1 liquid sample sensor that measures a range of 0.02–20 Wm/K. The instrument is based on the hot wire method, which measures heating and cooling cycles of 90 s at 1 s intervals. Measurements were performed in a 50 ml jacketed glass vessel connected to a thermoregulated water bath. The temperature was varied in a range of 303.15 to 333.15 K.

Densities were measured with a densimeter Anton Paar DMA 4500 M with an accuracy of 0.00005 g/cm<sup>3</sup> and repeatability of 0.00001 g/cm<sup>3</sup> in the temperature range of 293.15 to 363.15 K.

Viscosity measurement was performed using the Brookfield DV2T viscometer with a small sample adapter and the Spindle SC4-18 with an accuracy of ± 1.0 percentage and repeatability of ± 0.2percentage. Measurements were carried out with the help of Reocalc T software, and measurements were taken every 10 degrees in the temperature range of 303.15 to 353.15 K.

Ionic liquids were dried at 343.15 K under vacuum in the presence of phosphorus pentoxide for 36 h. After this treatment, water content was determined in triplicate using the Karl-Fischer titration method, using Metrohm 870 KF Titrino Plus.

The preparation of mixtures was performed by combining equimolar amounts of each IL. Each mixture was heated to 323.15 K and stirred subsequently for 12 h at room temperature and was again dried by the same procedure used before for pure ionic liquids.

For the preparation of the Ionanofluids, four weight fractions, 0.04 wt%, 0.06 wt%, 0.08 wt%, and 0.1 wt%, were considered. Each sample was prepared in two steps. First, the calculated amount of base fluid was divided into two portions, and the MWCNTs were dispersed in the first half of each ionic liquid mixture. Next, it was mixed well for 15 min with a magnetic stirrer and then sonicated

in an ultrasonic bath for two h. Subsequently, the second amount of the base fluid was added, and each sample was stirred for another 15 min. Finally, they were sonicated for another 4 h, obtaining the desired Ionanofluids.

### 3. Results and discussion

#### 3.1. Selection of ionic liquid mixtures and elaboration of Ionanofluids

From a previous work carried out by this group, in which the thermophysical properties of fifteen mixtures of ionic liquids were evaluated, two equimolar mixtures, [Omim][PF<sub>6</sub>] + [Odmim][PF<sub>6</sub>] and [Omim][PF<sub>6</sub>] + [Odmim][BF<sub>4</sub>], called ILM1 and ILM2 respectively, shown an increase in heat capacity concerning pure ILs. ILM1 and ILM2 were mainly BF because they differ in one anion. The goal is to evaluate the influence of these ions on the thermo-physical properties of both INF.

On the other hand, considering that an INF is a thermodynamically stable dispersion, we have studied INFs composed of MWCNTs with low weight percentages of 0.04, 0.06, 0.08, and 0.1 wt%. MWCNT suspensions at these concentrations are easily dispersed with a bath sonicator. Visual analysis observed that the suspensions did not exhibit aggregates and were stable for at least three months, although this stability depends on the sonication time. Fig. 2 shows a visual comparison of three INFs with 0.1 wt% nanotubes dispersed at different sonication times. The highest stability even after three months is obtained with a sonication time of 4.5 h.

#### 3.2. Heat capacity

The heat capacity ( $C_p$ ) is one of the main properties that must be analyzed in an HTF, and it is essential to know the influence of nanoparticles in the INFs. The heat capacity of the INFs were determined by DSC, and it is calculated from the following equation (1) [20]:

$$C_p = \frac{1}{m} \frac{\partial H}{\partial T_p} = \frac{1}{m} \frac{\partial H / \partial t}{\partial T / \partial t_p} = \frac{1}{m} \frac{\Delta P}{\beta} \quad (1)$$

where  $m$  is the sample mass,  $H$  is enthalpy,  $\Delta P$  is the absolute value of the heat flow to the sample, and  $\beta$  is the heating rate described by equation (2):

$$\beta = \frac{T_t - T_0}{t} \quad (2)$$

where  $T_0$  is the initial temperature, and  $T_t$  is the temperature at time  $t$ . A quadratic polynomial equation (3) fits the heat capacity data.. The whole data is available in the [Supplementary Material](#).

$$C_p = A_0 + A_1 \cdot T + A_2 \cdot T^2 \quad (3)$$

The heat capacities of the BF and the INFs with 0.04, 0.06, 0.08 and 0.1 wt% as a function of temperature are shown in Fig. 3.

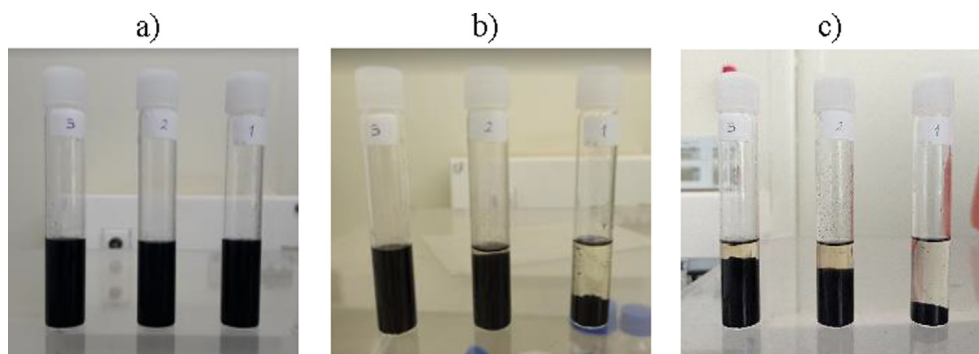
Fig. 3a shows as the heat capacity of ILM1 + MWCNTs increases with temperature, but INFs have lower  $C_p$  values respect to BF. ILM2 + MWCNTs (Fig. 3b) also presents a rise in heat capacity with increasing temperature, but unlike ILM1, heat capacity increases above the BF. The most remarkable enhancement was observed at 0.1 wt%, at which  $C_p$  increases by approximately 200 % at 373.15 K.

This favorable and surprising result could be attributed to the different influences exerted by [PF<sub>6</sub>]<sup>−</sup> and [BF<sub>4</sub>]<sup>−</sup> anions. However, the influence of these anions in a mixture of ionic liquids differs from the behavior observed by other researchers. Table 1 compares our results with those obtained by other researchers, who also used [PF<sub>6</sub>]<sup>−</sup> and [BF<sub>4</sub>]<sup>−</sup> anions in their base fluids. The variation of  $C_p$  above or below the BF values does not seem to follow a pattern related to the presence of these anions. The results obtained contradict the conclusion made by Oster et al. [21], who after evaluating different types of ionic liquids with MWCNTs, boron nitride and graphite, stated that the increase in heat capacity is determined by the type of nanoparticles, rather than the type of ionic liquid.

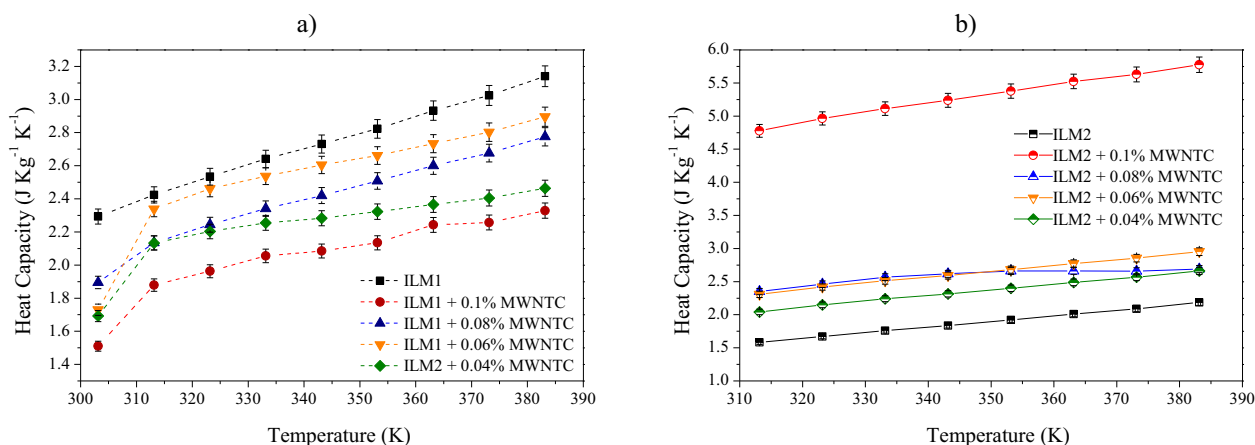
Nanoscale interactions between the nanotube surface and the ions that make up the BF could explain the variation of  $C_p$  in INFs. After the attempts made to understand the behavior of ILs on the surface of CNTs [25–32], the formation of interfacial nano-layers around the nanotubes is the idea that has gained the most support in recent years to explain the increase in heat capacity.

Molecular dynamics (MD) has been a valuable tool to understand that a complex of intermolecular  $\pi$ - $\pi$  stacking, van der Waals forces, electrostatic forces, and hydrogen bonds facilitate imidazolium cations and anions interaction with carbon nanotubes, forming monolayers or multilayers [33] in perpendicular [34,35] or parallel arrangement. [33]

There are limited precedents related to the BFs used in this study. Frolov et al. [35] simulated and demonstrated that [Omim]<sup>+</sup> cations orientated parallel to the CNT surface. A recent and previous study on the analysis of ionic liquid mixtures in INFs [36] investigated the dynamics and hydrogen bonding of an ionic liquid equimolar mixture of ethyl-3-methylimidazolium tetrafluoroborate ([Emim][BF<sub>4</sub>]) and 1-butyl-3-methylimidazolium hexafluorophosphate ([Bmim][PF<sub>6</sub>]) and single-walled carbon nanotubes (CNTs) of different diameters. The results revealed that the hydrogen bonds formed by [BF<sub>4</sub>]<sup>−</sup> and [PF<sub>6</sub>]<sup>−</sup> anions around CNTs exhibit competitive behavior. The hydrogen bonds between [PF<sub>6</sub>]<sup>−</sup> anions and imidazolium cations are higher in quantity and more substantial than those between [BF<sub>4</sub>]<sup>−</sup> anions as the CNT diameter increases, and although both are strong in the first solvation layer,



**Fig. 2.** a) INFs freshly sonicated for 1 h (1), 3 h (2), and 4.5 h (3); and the same b) INFs sonicated for 1 h (1), 3 h (2), and 4.5 h (3) after three months, c) INFs sonicated for 1 h (1), 3 h (2), and 4.5 h (3) after 1 year.



**Fig. 3.** Variation of heat capacity in function of temperature. a) [Oim][PF<sub>6</sub>] + [Odmim][PF<sub>6</sub>] (ILM1) + MWNTC; b) [Oim][PF<sub>6</sub>] + [Odmim][BF<sub>4</sub>] (ILM2) + MWNTC.

**Table 1**

Comparison of the variation of heat capacity obtained in other studies.

| Reference              | NP       | wt%           | BF                                                  | Cp       |
|------------------------|----------|---------------|-----------------------------------------------------|----------|
| Castro et al. [5]      | MWCNT    | 1, 1.5        | [Bmim][PF <sub>6</sub> ]                            | increase |
| Wang et al. [22]       | MWCNT    | 0.03, 0.06    | [Hmim][BF <sub>4</sub> ]                            | decrease |
| Wang et al. [22]       | Graphene | 0.03, 0.06    | [Hmim][BF <sub>4</sub> ]                            | decrease |
| Shevelyova et al. [23] | MWCNT    | 12.4          | [Bmim][BF <sub>4</sub> ]                            | decrease |
| Shevelyova et al. [23] | MWCNT    | 11.4          | [Bmim][PF <sub>6</sub> ]                            | decrease |
| Liu et al. [24]        | Graphene | 0.03, 0.06    | [Hmim][BF <sub>4</sub> ]                            | decrease |
| Oster et al. [21]      | MWCNT    | 0.5, 1.0, 3.0 | [Hmim][PF <sub>6</sub> ]                            | increase |
| This work              | MWCNT    | 0.1           | [Oim][PF <sub>6</sub> ] + [Odmim][PF <sub>6</sub> ] | decrease |
| This work              | MWCNT    | 0.1           | [Oim][PF <sub>6</sub> ] + [Odmim][BF <sub>4</sub> ] | increase |

there is a significant aggregation of [PF<sub>6</sub>]<sup>-</sup> ions around CNTs. Additionally, competitive interactions result in slower rotations of both anions and a more pronounced orientational distribution of both imidazolium cations in the first solvation layer around CNTs.

This preliminary background allows predicting that the forces affecting the ions around the CNTs influence the thermal properties, generating a variable interfacial thermal resistance based on the movement of the ions and the strength of the intermolecular forces. Thus, the existing competitiveness of the hydrogen bonds in the mixture with [BF<sub>4</sub>]<sup>-</sup> and [PF<sub>6</sub>]<sup>-</sup> ions could positively influence the heat capacity of the INFs, however, further studies on this subject should be done.

### 3.3. Thermal conductivity

Fig. 4 shows the thermal conductivity ( $k$ ) of BFs and INFs with 0.04, 0.06, 0.08 and 0.1 wt% as a function of temperature. Fig. 4a and 4c show that, in the studied temperature range (293.15 K to 333.15 K) at 0.1 MPa, temperature exerts a negligible influence on the thermal conductivity of BFs, remaining almost constant as temperature increases. At 293.15 K, INFs exhibit a higher thermal conductivity than BFs, being 1.5% and 2% higher in the case of ILM1 and ILM2, respectively. Nevertheless, there is a slight decrease of  $k$  with increasing temperature. This behavior is evidenced by observing the deviation of the thermal conductivities of the INFs concerning the BF ( $(K_{INF}-K_{BF})/K_{BF}$ ) in Fig. 4b and 4d.

The standard deviation associated with the thermal conductivity measurements by triplicate are consigned as the error bars in Fig. 4a and 4c, and considering the estimated error of  $\pm 0.005$  W/mK associated with the transient hot wire method [37], it might suggest that the addition of MWCNTs does not modify the thermal conductivity of the studied BFs, as in the case of other systems previously reported [22,38,39].

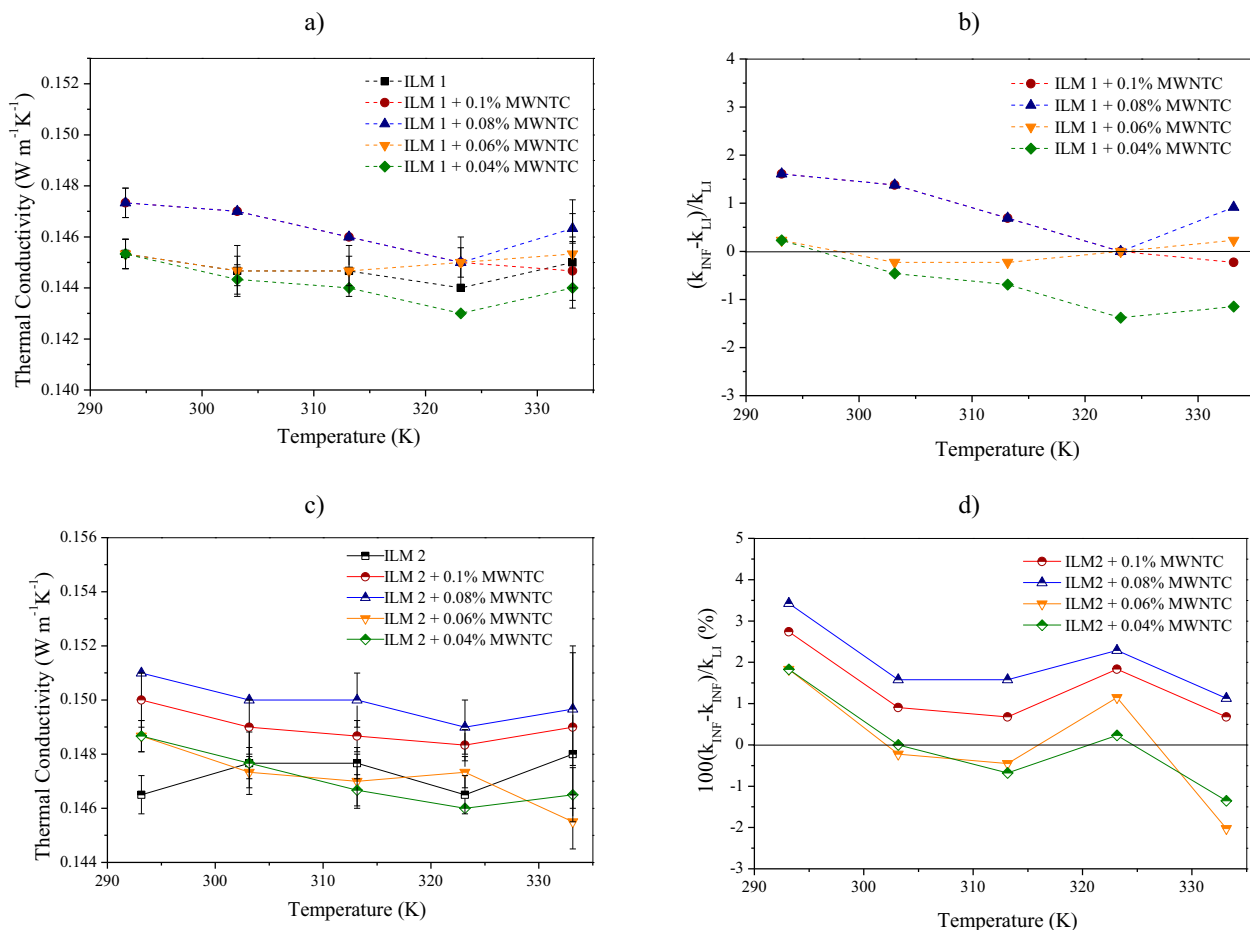
In this context, the weak influence of MWNTC on thermal conductivity could be associated with low concentrations. Thus, an increase in the weight percentages of MWCNTs would probably increase the thermal conductivity of the INFs. [40,41] The addition of small concentrations of nanomaterials on heat transfer fluids may cause essential variations in the thermal conductivity of these fluids. The extent of such variations depends on several factors, such as the chemical composition and physical properties of the base fluid and the nanomaterials, including their physical chemistry interactions. Furthermore, the thermal conductivity of the obtained INFs obeys complex mechanisms such as micro-convection, Brownian motion, and thermal diffusion in nanomaterials; that is why predicting the thermal conductivity improvement by a single model is still a challenge. For instance, models such as Hamilton and Crosser [42] and Murshed et al. [43] overvalue the thermal conductivity of ionanofluids based on multi-wall carbon nanotubes [44].

### 3.4. Density

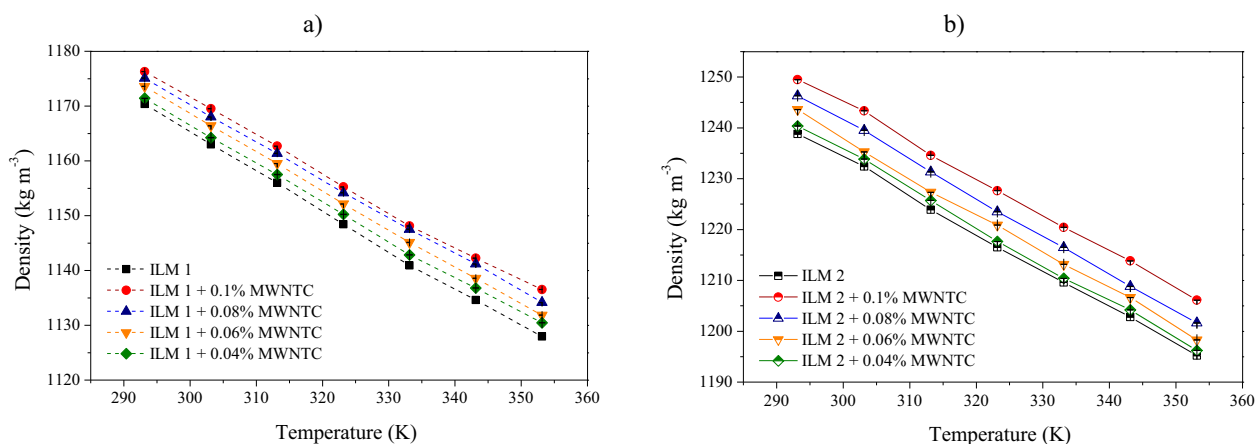
The density was measured at atmospheric pressure in the temperature range of 293.15 to 353.15 K for all INFs and mixtures. The obtained temperature dependent values were fitted with the polynomial Eq. (4), and the fitting parameters A0, A1, and A2 for temperatures  $T$  in K are presented in the Supplementary Material.

$$\rho = A0 + A1 \cdot T + A2 \cdot T^2 \quad (4)$$

According to other sources [45,46], the density of all INFs increased with nanoparticle concentration and decreased with temperature. Fig. 5 shows that in the case of the INFs under study, only a slight increase in density was observed with concentration, increasing to close to 1% for the concentration of 0.1% by weight.



**Fig. 4.** Variation of thermal conductivity in function of temperature. a) [Omim][PF<sub>6</sub>] + [Odmim][PF<sub>6</sub>] (ILM1) + MWNTC; b) [Omim][PF<sub>6</sub>] + [Odmim][PF<sub>6</sub>] (ILM1) + MWNTC respect to ILM1; c) [Omim][PF<sub>6</sub>] + [Odmim][BF<sub>4</sub>] (ILM2) + MWNTC; d) [Omim][PF<sub>6</sub>] + [Odmim][BF<sub>4</sub>] (ILM2) + MWNTC respect to ILM2.



**Fig. 5.** Variation of density in function of temperature. a) [Omim][PF<sub>6</sub>] + [Odmim][PF<sub>6</sub>] (ILM1) + MWNTC; b) [Omim][PF<sub>6</sub>] + [Odmim][BF<sub>4</sub>] (ILM2) + MWNTC.

### 3.5. Viscosity

The viscosity curves can be described by the Vogel-Fulcher-Tammann (VFT) equation (5) [13], where  $\mu_0$ ,  $\beta$ , and  $T_0$  are the correlation coefficients that are presented in the [Supplementary Material](#).

$$\mu = \mu_0 \cdot \exp\left(\frac{\beta}{T - T_0}\right) \quad (5)$$

**Fig. 6** shows the influence of temperature and MWNTC concentration on viscosity. It can be seen that viscosity is initially higher with MWNTC presence, especially in ILM1-based INFs. However, in all the cases, de viscosity decreases exponentially with increasing temperature. As observed by other authors [47–49], the decrease in viscosity with temperature is typical behavior to many INFs. Increasing temperature weakens the shear thinning behavior of



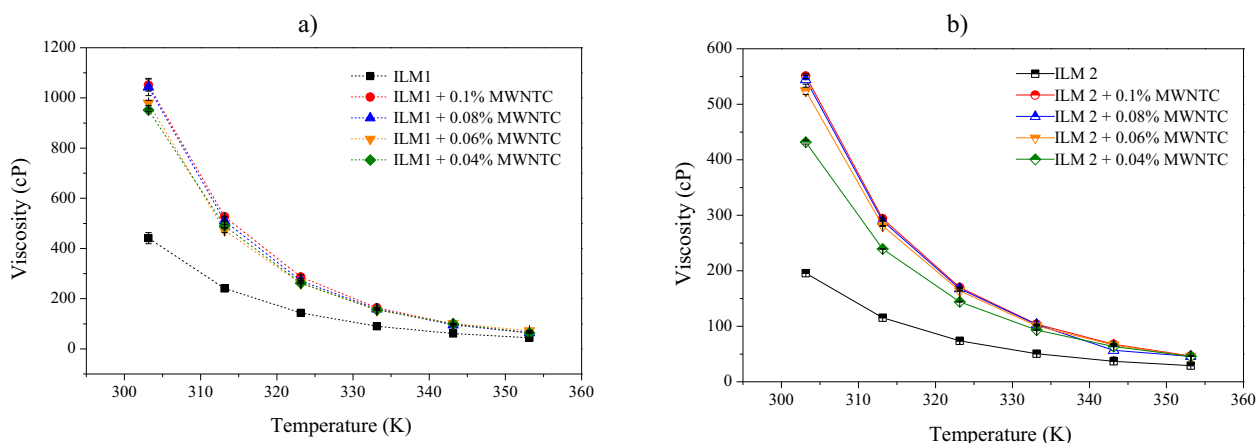


Fig. 6. Variation of viscosity in function of temperature. a) [Omim][PF<sub>6</sub>] + [Odmim][PF<sub>6</sub>] (ILM1) + MWNTC; b) [Omim][PF<sub>6</sub>] + [Odmim][BF<sub>4</sub>] (ILM2) + MWNTC.

INFs; therefore, the incorporation of carbon nanotubes does not cause a noticeable change in viscosity at working temperatures (above 500 K).

### 3.6. Thermal storage density

Thermal energy storage density,  $E$ , is a crucial parameter in a thermal energy storage system since it allows quantifying the energy stored by a material in a determined volume.

$E$ 's of the INFs and BF<sub>4</sub> were calculated with equation (6) [50,51]:

$$E = \rho \cdot C_p \cdot (T_o - T_i) \quad (6)$$

where  $T_o$  and  $T_i$  are outlet and inlet temperature, respectively,  $\rho$  is the density value kg/m<sup>3</sup>, and  $C_p$  is the heat capacity in kJ/kg·K. In order to compare the results at operational temperatures, a difference of 100 K ( $\Delta T$ ) has been used. The obtained thermal storage densities from the preliminary analysis, and those of other ionic liquids previously analyzed are compared in Table 2.

In order to evaluate INFs as a storage and heat transfer fluid, it is pertinent to compare conventional materials commonly used in solar energy storage, such as nitrate salts, Therminol VP-1, and mineral or silicone oil [50–52]. In this aspect, it is necessary to consider the operating temperatures of each material. As can be observed in Table 3, the thermal storage densities of the base fluids of the INFs are substantially higher than the values of the thermal oils, mainly due to the higher density and heat capacity of the mixtures. However, similar to what was observed with the  $C_p$  values, MWNTC decreases the ILM1 energy storage. Nevertheless,

ILM2 + 0.1% of MWCNT is favored, presenting an  $E$  value even higher than that of nitrate salts.

Considering the  $E$  values, this latter INF could be a good candidate as HTF in solar energy storage. However, further analysis is needed.

### 3.7. Thermal diffusivity

Thermal diffusivity ( $\alpha$ ) is the ability of a material to conduct thermal energy relative to its ability to store thermal energy. [53] The thermal diffusivity,  $\alpha$ , is defined by equation 6:

$$\alpha = \frac{k}{\rho \cdot C_p} \quad (6)$$

Using the results obtained in the previous analyses, we compared the variation of diffusivity as a function of temperature for Therminol VP-1, Nitrated Salts and the INFs with the highest  $C_p$ , ILM2 + 0.1% MWCNT, in a temperature range of 533.15 K–673.15 K.

Fig. 7 highlights a significant difference between the Nitrated Salts and the other two thermal fluids, with the lowest values for the INF.

Low thermal diffusivity implies low heat conduction in non-stationary systems, being a drawback for heat transfer but, at the same time, an advantage in terms of heat storage. In this term, the resulting INFs diffusivity is consistent with previously obtained energy storage density results and reinforces the idea of its possible use as a thermal storage material.

Table 2

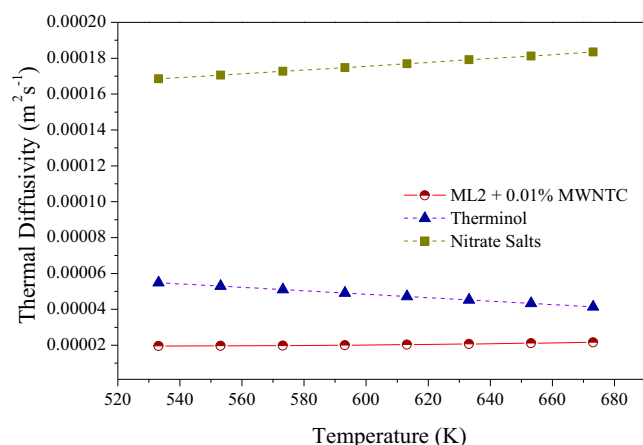
Comparison of thermal storage density of ionic liquids and INFs ( $\Delta T = 100$  K).

| Fluid                                                             | Reference | $C_p$ (kJ/kg·K) | $\rho$ (kg/m <sup>3</sup> ) | $E$ (MJ/m <sup>3</sup> ) |
|-------------------------------------------------------------------|-----------|-----------------|-----------------------------|--------------------------|
| [Hmim][Cl]                                                        | [50]      | 1.9             | 1030                        | 170                      |
| [Hmim][PF <sub>6</sub> ]                                          | [50]      | 1.6             | 1300                        | 210                      |
| [Bmim][NTf <sub>2</sub> ]                                         | [50]      | 1.2             | 1440                        | 180                      |
| [Bmim][BF <sub>4</sub> ]                                          | [50]      | 1.7             | 1175                        | 200                      |
| [N <sub>444H</sub> ][NTf <sub>2</sub> ]                           | [51]      | 1.6             | 1241                        | 200                      |
| [C <sub>12</sub> mim][NTf <sub>2</sub> ]                          | [51]      | 1.6             | 1189                        | 177                      |
| [Omim][PF <sub>6</sub> ]                                          | [19]      | 1.6             | 1181                        | 189                      |
| [Odmim][BF <sub>4</sub> ]                                         | [19]      | 1.8             | 1190                        | 214                      |
| [Odmim][PF <sub>6</sub> ]                                         | [19]      | 2.6             | 1088                        | 283                      |
| [Omim][PF <sub>6</sub> ] + [Odmim][PF <sub>6</sub> ]              | [19]      | 2.9             | 1116                        | 324                      |
| [Omim][PF <sub>6</sub> ] + [Odmim][BF <sub>4</sub> ]              | [19]      | 2.3             | 1199                        | 276                      |
| [Omim][PF <sub>6</sub> ] + [Odmim][PF <sub>6</sub> ] + 0.1% MWNTC | This work | 2.3             | 1128                        | 259                      |
| [Omim][PF <sub>6</sub> ] + [Odmim][BF <sub>4</sub> ] + 0.1% MWNTC | This work | 5.6             | 1197                        | 670                      |

**Table 3**

Comparison of thermal storage density of ionic liquids mixtures, INFs, and common thermal fluids.

| Fluid                                                             | Reference | $\rho$ (kg/m <sup>3</sup> ) | $C_p$ (kJ/kg K) | T (K)<br>$T_{inlet} - T_{outlet}$ | $E$ (MJ/m <sup>3</sup> ) |
|-------------------------------------------------------------------|-----------|-----------------------------|-----------------|-----------------------------------|--------------------------|
| Mineral oil                                                       | [50]      | 770                         | 2.60            | 473.15–573.15                     | 200                      |
| Silicone oil                                                      | [50]      | 900                         | 2.10            | 573.15–673.15                     | 189                      |
| Synthetic oil                                                     | [50]      | 900                         | 2.30            | 523.15–533.15                     | 207                      |
| Therminol VP-1                                                    | [52]      | 991                         | 1.77            | 523.15–663.15                     | 246                      |
| [Omim][PF <sub>6</sub> ] + [Odmim][PF <sub>6</sub> ]              | This work | 1116                        | 2.97            | 523.15–663.15                     | 464                      |
| [Omim][PF <sub>6</sub> ] + [Odmim][BF <sub>4</sub> ]              | This work | 1199                        | 2.28            | 523.15–663.15                     | 383                      |
| [Omim][PF <sub>6</sub> ] + [Odmim][PF <sub>6</sub> ] + 0.1% MWNTC | This work | 1128                        | 2.27            | 523.15–663.15                     | 358                      |
| [Omim][PF <sub>6</sub> ] + [Odmim][BF <sub>4</sub> ] + 0.1% MWNTC | This work | 1197                        | 5.65            | 523.15–663.15                     | 945                      |
| Nitrate salts                                                     | [50]      | 1870                        | 1.54            | 538.15–838.15                     | 864                      |

**Fig. 7.** Comparison of the variation of diffusivity as a function of temperature for Nitrated Salts, Therminol, and INF.

#### 4. Conclusions

The purpose of the current study was to make an additional contribution to research on new energy transfer and storage fluids.

Two equimolar mixtures of ionic liquids, [Omim][PF<sub>6</sub>] + [Odmim][PF<sub>6</sub>] (ILM1) and [Omim][PF<sub>6</sub>] + [Odmim][BF<sub>4</sub>] (ILM2), were used as base fluids in the preparation of Ionanofluids composed of MWCNTs with weight percentages of 0.04, 0.06, 0.08 and 0.1 wt%.

Thermophysical properties such as heat capacity, thermal conductivity, density, and viscosity were measured as a function of temperature and MWCNT weight fraction to evaluate which fluids enhanced their properties for solar storage applications.

Substantial differences were observed between the INFs, which are mainly associated with the interaction between anions and carbon nanotubes. In the case of  $C_p$ , the addition of 0.1 wt% MWCNT to the ILM1 base fluid decreased the heat capacity from 2.9 kJ/kg·K to 2.3 kJ/kg·K at 373.15 K, while ILM2 increased from 2.3 kJ/kg·K to 5.6 kJ/kg·K.

In contrast, no significant changes in thermal conductivity were observed concerning the base fluids, while slight increases in density were observed under the concentration of each INF, reaching close to 1% for the cases of greater concentration. Similarly, the viscosity was not modified by the incorporation of carbon nanotubes in the working temperature range (above 500 K) because it decreases with increasing temperature in each of the cases.

Finally, a comparison with conventional materials commonly applied for thermal energy storage was performed. Thermal storage density ( $E$ ) and thermal diffusivity ( $\alpha$ ) values were calculated, and in the case of ILM2 + 0.1% MWCNT an  $E$  value of 945 MJ/m<sup>3</sup> higher than Nitrated Salts and Therminol VP-1 was observed. The

$\alpha$  values obtained were much lower than conventional fluids, which demonstrated the high capacity to store energy of the mixture. However, it is necessary to perform additional studies to clearly define their possible use as energy transfer and storage fluids.

#### CRediT authorship contribution statement

**Jenifer Cavieres:** Investigation, Data curation, Validation. **María José Inestrosa-Izurieta:** Conceptualization, Visualization, Writing – review & editing. **Diego A. Vasco:** Resources, Visualization, Writing – review & editing. **Julio I. Urzúa:** Conceptualization, Methodology, Investigation, Project administration, Writing – original draft, Writing – review & editing.

#### Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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#### Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.molliq.2022.118677>.

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